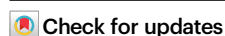


Lattice expansion of hybrid perovskite inhibits halogen interstitial generation and enhances solar cell performance

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Tongtong Lu¹, Zhiya Dang^{1,2}✉, Yuqing Luo¹, Yang Li¹, Xiaobin Rao¹, Zhe Zhang¹, Yuecheng Hu¹, Zelong Chen¹, Zihao Li¹, Urko Petralanda³✉, Jiangsheng Xie^{1,2}, Jiancan Yu^{1,2} & Pingqi Gao^{1,2}✉

One of the primary challenges for halide perovskite technology lies in its instability. Water is a ubiquitous environmental factor known to induce lattice expansion to perovskite. Lattice expansion has been proven beneficial to perovskite solar cell, but little is known about its role in lattice stability. In this work, we expose perovskites with pre-adsorbed water to vacuum and investigate the evolution of their structure. Our results show that vacuum induces water desorption and results in the removal of lattice expansion. This lattice recovery leads to an increasing lattice distortion, ultimately favoring the formation of halogen trimers in the case of hybrid perovskites. We thus find that the lack of lattice expansion facilitates the generation of defects detrimental to hybrid perovskite devices. Here, we report the crucial role of lattice expansion in preventing degradation of hybrid halide perovskites, which is of critical importance in the context of solar cell fabrication.

Metal halide perovskites have stood out as a central focus in modern semiconductor optoelectronics. This is owing to their superior properties such as long carrier diffusion length, high absorption coefficient and defect tolerance^{1–9}. Due to the soft lattice nature, the optoelectronic properties of perovskite are highly sensitive to the lattice volume. Earlier works revealed that external stimulus, such as continuous light illumination, can induce slight lattice expansion to perovskite, which leads to a significant decrease in the non-radiative recombination coefficient of defects and enhances the device performance^{10–12}.

The interaction of perovskite with water, a most common environmental factor, has been studied intensively. Water is often unintentionally adsorbed during perovskite characterization and device fabrication, subsequently causing lattice expansion of the perovskite surface^{13,14}. The pre-adsorbed water can also be eliminated through processes such as heating and vacuum exposure. In particular, perovskite is inevitably exposed to high vacuum throughout the fundamental research and space applications¹⁵.

Earlier, a number of studies were dedicated to investigating the structural changes of perovskites under light illumination or heating based on vacuum-based techniques such as X-ray photoelectron spectroscopy (XPS)^{16,17}. Nevertheless, the potential role of the vacuum in removing the pre-adsorbed water and the possibly induced structural changes was not fully considered in these works. It still remains uncertain how the elimination of water, and its concomitant removal of lattice expansion, affects the structure and property of perovskite, as well as the performance of the device.

Pioneering efforts showed that prolonged vacuum exposure can cause the instability of perovskite-based devices^{18,19}. There are strong indications suggesting that the elimination of lattice expansion may be an important origin of the instability of the devices^{18,19}, but a convincing confirmation of this is lacking to date. Therefore, it is imperative to explore the structural evolution of halide perovskites upon recovery of lattice expansion, as it is not only important for understanding the intrinsic structure of perovskite, but also crucial for preventing the degradation of perovskite.

¹School of Materials, Sun Yat-sen University, Shenzhen, China. ²State Key Laboratory of Optoelectronic Materials and Technologies, Sun Yat-sen University, Guangzhou, China. ³Department of Applied Mathematics, University of the Basque Country UPV/EHU, Vitoria-Gasteiz, Spain.

✉ e-mail: dangzhy3@mail.sysu.edu.cn; urko.petralanda@ehu.eus; gaopq3@mail.sysu.edu.cn

In this work, we first present a direct observation of structural and optical evolution conducted on MAPbBr₃ crystal with pre-adsorbed water during vacuum exposure for a time scale of hours. These experiments allow us to identify two major structural changes: firstly, vacuum exposure induces water desorption, causing lattice recovery through the elimination of lattice expansion; secondly, this removal of lattice expansion further leads to the generation of halogen trimers. These structural progressions can be correlated well with the luminescence changes observed during vacuum exposure. We propose that the emergence of trimers after water desorption occurs through an intermediate phase (60° phase), wherein MA⁺ molecules are rotated 60° around the C–N bond axis with respect to the ground state. To support it, we do density functional theory (DFT) calculations in combination with Raman spectroscopy. The calculations suggest that the 60° structure is metastable and extremely close in energy to the ground state at equilibrium volume, albeit prohibitive at expanded lattices. In addition, the 60° phase presents a shorter Br–Br distance and a weaker bonding network around the Br atoms, facilitating the thermally assisted generation of Br₂ dimers. Comparison of DFT and Raman results shows a redshift of a C–N stretching mode peak only present in the 60° phase, which can serve as a potential fingerprint of its presence in this intermediate phase in the material. The enhanced presence of dimers can further trigger the formation of bromine trimers upon interaction with a third bromide ion. The evolution trend of MAPbBr₃ also applies to candidate hybrid perovskites for photovoltaic applications, such as FAPbI₃ and Cs_{0.05}MA_{0.1}FA_{0.85}PbI₃ films. The defect generated due to the lack of lattice expansion constitutes a non-radiative recombination channel of the carrier, which is found to deteriorate the solar cell performance. In contrast, the water desorption and the contaminant lattice restoration do not lead to the formation of halogen trimers in fully inorganic perovskites such as CsPbBr₃, emphasizing the essential role of organic cation in the instability of hybrid perovskite. These findings are not only essential for understanding the intrinsic structure and fundamental properties of such materials, but also provide inspiring insights for optimization of relevant device performance and stability.

Results

We first use MAPbBr₃ as a model material to study the structural evolution induced by the water desorption process under vacuum exposure. The MAPbBr₃ single crystals are synthesized using the inverse temperature crystallization (ITC) method²⁰, for which the procedure is included in the Supplementary Information (SI). The as-synthesized crystal shows high crystallinity with a smooth surface, as indicated by the UV-vis absorption spectra and inset images in Fig. 1a & Supplementary Fig. 1. X-ray diffraction (XRD) of the crystal after grinding matches well with the cubic phase of MAPbBr₃ as shown in Fig. 1b. Different batches of crystals have comparable quality. (Supplementary Fig. 2) The as-synthesized crystal and the powder obtained from grinding are first exposed to ambient atmosphere with relative humidity (RH) of 70% at 1 atm for 3 h before transferring to vacuum for further experiment.

PL evolution

In situ photoluminescence (PL) measurements are conducted on the MAPbBr₃ single crystal in a cryostat chamber equipped with vacuum (pressure = 1×10^{-6} mbar). The PL measurement configuration is shown in Supplementary Fig. 3. Varying-temperature PL is conducted after vacuum exposure for 0 h, 5 h, 16 h, 22 h and 48 h, respectively. The excitation light is off between PL capturing, and such light illumination condition does not cause obvious PL change as confirmed in Supplementary Fig. 4. Figure 1c–e reports the PL contour of the crystal at 0 h, 5 h and 16 h. The PL spectra at 0 h exhibit dual emission peaks at low temperature as seen in Fig. 1f, which can be attributed to free exciton (FE) and bound exciton (BE) recombination^{21,22}. The emission peak

positions change abruptly at -100 K due to the occurrence of a structural transition between the tetragonal and orthorhombic phase²³. (Fig. 1g)

After 5 h vacuum exposure, the dual emission peaks exhibit an obvious redshift of 0.02 eV, as shown in the low-temperature PL spectra in Fig. 1f. The redshift becomes negligible with increasing temperature, as reported in Supplementary Fig. 5. The PL redshift at low temperatures indicates vacuum-induced water desorption and contaminant lattice compression, which leads to a smaller bandgap. At higher temperatures, the electron-phonon coupling effect becomes larger and the bandgap is less sensitive to lattice volume²⁴. Besides, after 5 h vacuum exposure, the sudden change of PL peak position at -100 K is not observed, indicating that the phase transition has been inhibited due to the lattice shrinkage. (Fig. 1d, g)

With the vacuum exposure extended up to 16 h, a broad sub-bandgap emission emerges at 2.05 eV at low temperatures, as seen in Fig. 1e. Simultaneously, the dual peak positions shift back partially, and the phase transition reverses back to that of 0 h (Fig. 1f, g). Even longer exposure of 22 h and 48 h results in further reversal of the dual peak position, suggesting the re-occurrence of lattice expansion. (Fig. 1f) Moreover, the PL contour maps at 22 h and 48 h in Supplementary Fig. 6 show that the intensity of dual emission decreases while that of the vacuum-induced emission increases with further vacuum exposure, which indicates the defect formation. Apart from the above changes, the dual emission peak width increases at 5 h and reverses partially after 48 h exposure, as shown by the fitting parameters of three emission peaks listed in Supplementary Table 1. These observations imply that the initial structure with pre-adsorbed water has a reduced distortion owing to the lattice expansion. The lattice distortion becomes apparent after the water removal. During prolonged vacuum exposure, the lattice tends to expand again, probably due to the generated defect, thereby reducing the distortion.

To confirm water is the desorbed species causing the above change, we perform a control experiment in which several MAPbBr₃ samples are exposed to air with different humidity (RH = 10%, 30% and 80%). We track the PL after vacuum exposure for different durations (1×10^{-6} mbar, 0 h, 7 h, 25 h). Supplementary Fig. 7 shows that the sample treated with higher humidity experiences a more pronounced PL redshift and a higher defect emission peak intensity, confirming that the vacuum-induced changes are linked to water desorption. Note that this control experiment uses MAPbBr₃ polycrystalline film for which the detailed preparation procedure is included in the Methods section. The similar evolution of polycrystalline film indicates the generality of structural evolution among different forms of samples. Besides, a longer vacuum exposure duration is required for the water desorption-induced lattice shrinkage to manifest in the polycrystalline films, which is probably due to the tensile strain formed during the high-temperature annealing step of the synthesis.

The decay dynamics of the sub-bandgap emission are further explored by the time-resolved PL (TRPL) measurements. Figure 1h presents the TRPL spectra in three different energy bands corresponding to three emission peaks measured at 100 K on the MAPbBr₃ crystal after 48 h of vacuum exposure. The TRPL data can be well-fitted with a tri-exponential function (Supplementary Fig. 8), and an effective decay lifetime (τ_{eff}) is calculated based on the fitting results (Supplementary Table 2). The obtained τ_{eff} of vacuum-induced emission (7.25 μ s) is much longer than FE and BE emissions (0.23 μ s, 1.55 μ s), which is linked to the nature of the vacuum-induced defect (Br⁺ interstitials) and will be discussed further later.

To understand the origin of vacuum-induced sub-bandgap emission at 2.05 eV, we further perform power-dependent PL measurements on the MAPbBr₃ crystal after 48 h of vacuum exposure. Supplementary Fig. 9 presents the PL analysis under varying incident power density from 2.7 to 27.0 mW/cm² at a temperature of 100 K. The exponent (α) obtained from fitting of the emission intensity (I) against

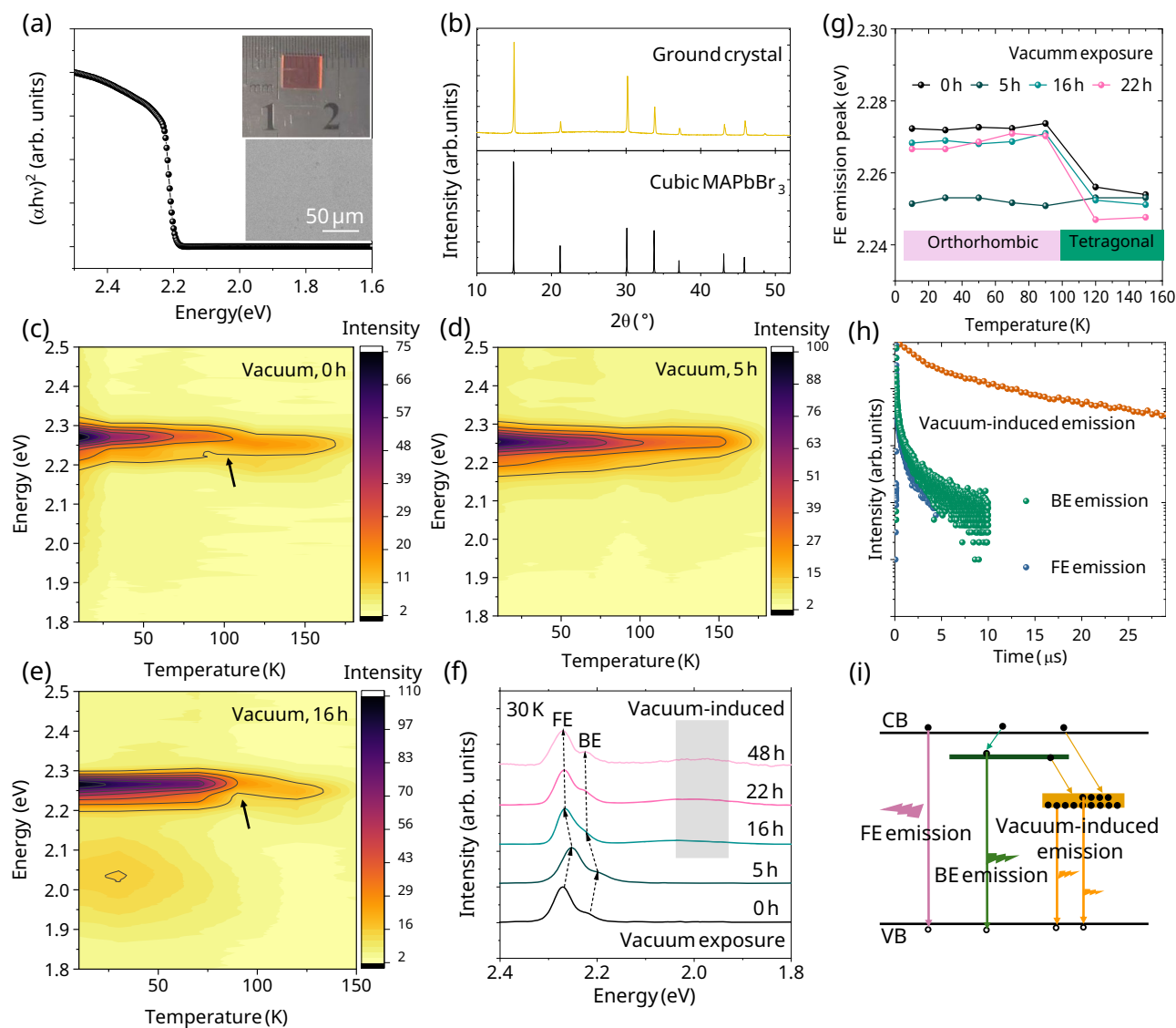


Fig. 1 | Characterization of as-synthesized MAPbBr₃ single crystal and vacuum-induced PL evolution. **a** Tauc plot fitting of UV-vis absorption spectra, the inset shows its optical and SEM image; **b** XRD of crystal after grinding, in comparison with the reference cubic phase; PL contour of the same crystal after vacuum exposure for **c** 0 h, **d** 5 h, and **e** 16 h under pressure of 1×10^{-6} mbar, in which the sudden change of peak position corresponds to phase transition, as labelled by the

black arrows; **f** normalized 30 K PL spectra and **g** variation of FE peak position with temperature after different vacuum exposure duration, inset shows the corresponding phase transition in the measured temperature range; **h** TRPL spectra in three energy bands corresponding to three emission peaks, measured at 100 K after 48 h vacuum exposure; **i** Sketch showing the origin of three emission peaks. Source data are provided as a Source Data file.

the power density (W) based on $I = I_0 W^\alpha$ is 1.29, 1.60, and 0.99 for the FE emission, BE emission and vacuum-induced emission respectively. This result suggests that the vacuum-induced emission stems from the radiative recombination of excitons, while the FE and BE emissions have contributions from both excitons and free carriers²⁵. Besides, the vacuum-induced emission peak position has an obvious blueshift from 1.88 eV to 1.98 eV with increasing power density, while the other two-peak positions remain unchanged. The blueshift can be attributed to the filling of the defect-related band with increasing power density and the resultant widening of the energy gap between the trap state and valence band, as illustrated by the sketch in Fig. 1i.

XRD evolution

Further, MAPbBr₃ samples with pre-adsorbed water are exposed to vacuum for different durations, and XRD analysis is used to unveil the vacuum-induced lattice variation. MAPbBr₃ samples of both single

crystal and powder obtained from grinding with pre-adsorbed water are first exposed to vacuum (pressure = 1×10^{-6} mbar) and then taken out to ambient briefly for XRD measurements between two adjacent vacuum exposures, as shown in Fig. 2a, b. Figure 2c and Supplementary Fig. 10 report the XRD of MAPbBr₃ powder and single crystal, respectively. The magnified XRD of various diffraction peaks is presented in Fig. 2d and Supplementary Fig. 11.

Diffraction peaks of both samples exhibit a two-peak feature, which cannot be attributed to any structure formed by phase transition of MAPbBr₃ as seen in Supplementary Fig. 12. Instead, this two-peak feature points to a slightly different structure at crystal surface compared to the interior. At 0 h, the lower angle branch of the two-peak feature is stronger, which can be attributed to the expanded lattice at the surface. After 5 h vacuum exposure, the intensity of this branch decreases significantly, suggesting the removal of water and restoration of the surface lattice. Accordingly, the higher angle branch

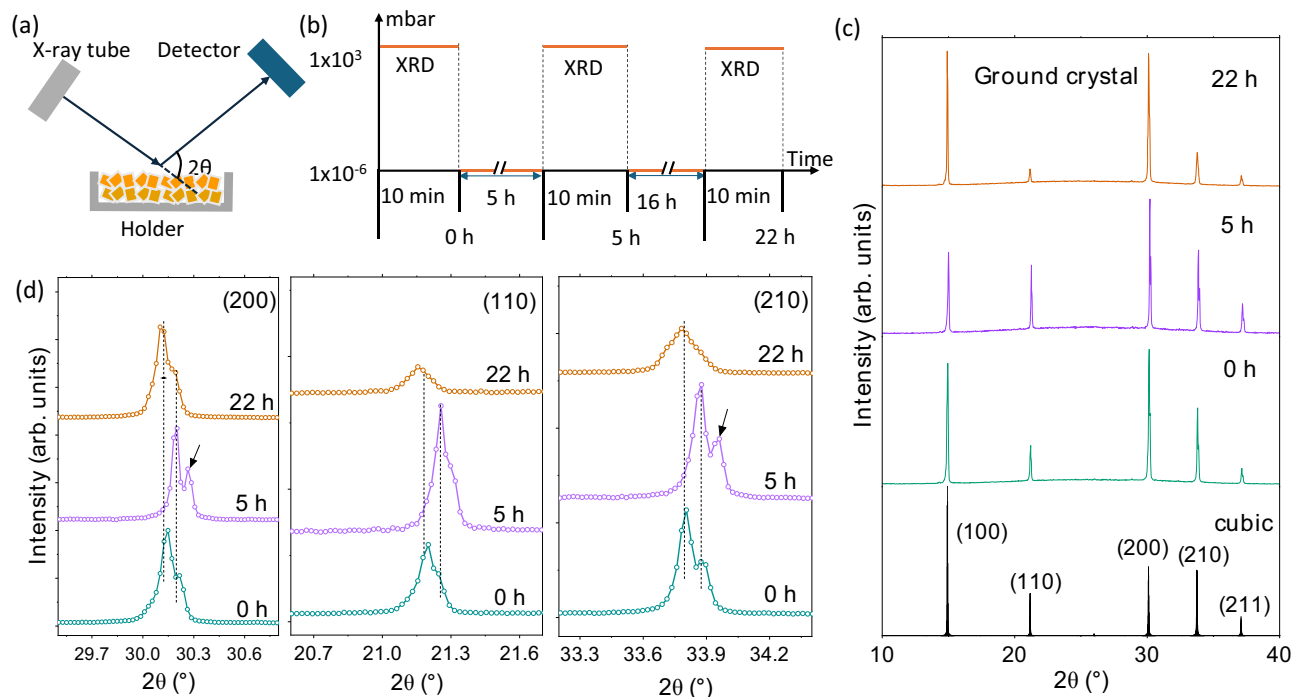


Fig. 2 | Evolution of MAPbBr₃ crystal structure with vacuum exposure.

a, b Schematics of the XRD measurement configuration and procedure; **c** XRD of the powder obtained from grinding, before and after vacuum exposure for 5 h and 22 h under pressure of 1×10^{-6} mbar, in comparison with the standard cubic phase. **d** Magnified XRD of (200), (110), and (210) diffraction peaks with a two-branch

feature shown by the dashed lines, in which two branches correspond to the structure of expanded lattice at surface and interior lattice respectively. The lower angle branch dominates at 0 h and 22 h, while the higher angle branch takes over at 5 h. The higher angle branch is accompanied by a shoulder at 5 h, as labelled by the black arrows. Source data are provided as a Source Data file.

representing the interior lattice structure dominates at 5 h. The two-branch feature of diffraction peak can also be observed in the XRD results of perovskites in various literatures^{26–28}, indicating the inevitable water adsorption occurring during sample transfer processes. Besides, the higher angle branch at 5 h is accompanied by a shoulder on the right side as labelled by the black arrows in Fig. 2d, probably due to the appearance of metastable 60° phase in lattice with absence of expansion as will be discussed further in the later sections. In the later stage of vacuum exposure at 22 h, the lower angle branch exhibits re-dominance, suggesting the re-occurrence of lattice expansion. Note that the diffraction peak splitting is not observed for powder with much smaller crystallite sizes because water adsorption induces lattice expansion efficiently in the entire crystallites. (Supplementary Fig. 12) Accordingly, the diffraction peak of samples with small crystallite sizes such as polycrystalline films exhibits a shift towards higher angle after short vacuum exposure and reverses back after longer exposure, as will be shown by the XRD results later.

Besides, the characteristic diffraction peaks of perovskite hydrate are not observed in our XRD analysis. This is further confirmed by the Fourier transform infrared spectroscopy (FTIR) analysis of moisture-exposed MAPbBr₃ sample, which does not show the characteristic peaks of the hydrate phase as reported in refs. 29,30. (Supplementary Fig. 13) These results indicate that the trace water molecules are adsorbed to the surface lattice and cause lattice expansion by residing at the relatively spacious interstices of large frame work composed of Pb and Br atoms, despite the presence of MA⁺ molecules³¹.

The vacuum-induced water desorption is reversible if the sample is exposed to ambient conditions again after the initial vacuum exposure. We monitor the XRD of 5 h vacuum-exposed MAPbBr₃ powder after taking it out to ambient condition (RH = 70%), which shows that the intensity of water adsorption-related diffraction peak increases in a time scale of 3 h (Supplementary Fig. 14). This indicates that the air exposure should be minimized in the ex-situ XRD measurement which

is the case of characterization procedure in Fig. 2b. Besides, we also attempt an in-situ structural analysis in an XRD instrument equipped with a vacuum. However, the low vacuum accessible in the instrument (pressure = 1×10^{-3} mbar) does not cause any structural variation even after 40 h exposure as shown in Supplementary Fig. 15. This result indicates that the water molecules are adsorbed onto perovskite crystal surface in the form of chemisorption, and the energy barrier for water molecules to leave perovskite surface lattice is relatively large due to the bonds formed between water-perovskite such as hydrogen bond formed between the O atoms of water molecules and H_N atoms of MA⁺ molecules³¹.

XPS evolution

To investigate the compositional changes during the vacuum exposure, we perform in-situ XPS measurements on a MAPbBr₃ single crystal after different vacuum exposure durations. The XPS spectra are captured on the crystal surface after 0 h, 2 h, 4 h, 7 h, 10 h, 12 h, 15 h, 19 h and 23 h exposure to the ultrahigh vacuum in XPS chamber (1.6×10^{-9} mbar). The crystal is grounded by wrapping an indium (In) foil around the edge, which is commonly used in XPS to avoid charge accumulation. The In foil has a certain distance from the measurement spot and does not affect the XPS measurement. The constant position of the O 1s peak in Fig. 3a indicates the absence of charging.

Figure 3b–f presents the Cl 1s, Pb 4f, N 1s, valence band (VB) and Br 3d XPS spectra after different vacuum exposure durations. The following changes can be observed from the XPS results: (1) The binding energy (BE) of all the peaks except O 1s remain unchanged until 4 h and starts shifting toward lower BE at 7 h until reaching minimum at 15 h, as plotted in Fig. 3g. It was reported earlier that the water exposure onto a cleaved MAPbBr₃ crystal surface causes all the XPS peaks except O 1s to shift toward higher BE, while with oxygen exposure the VBM shifts to lower BE²⁶. Our observation indicates that the exact

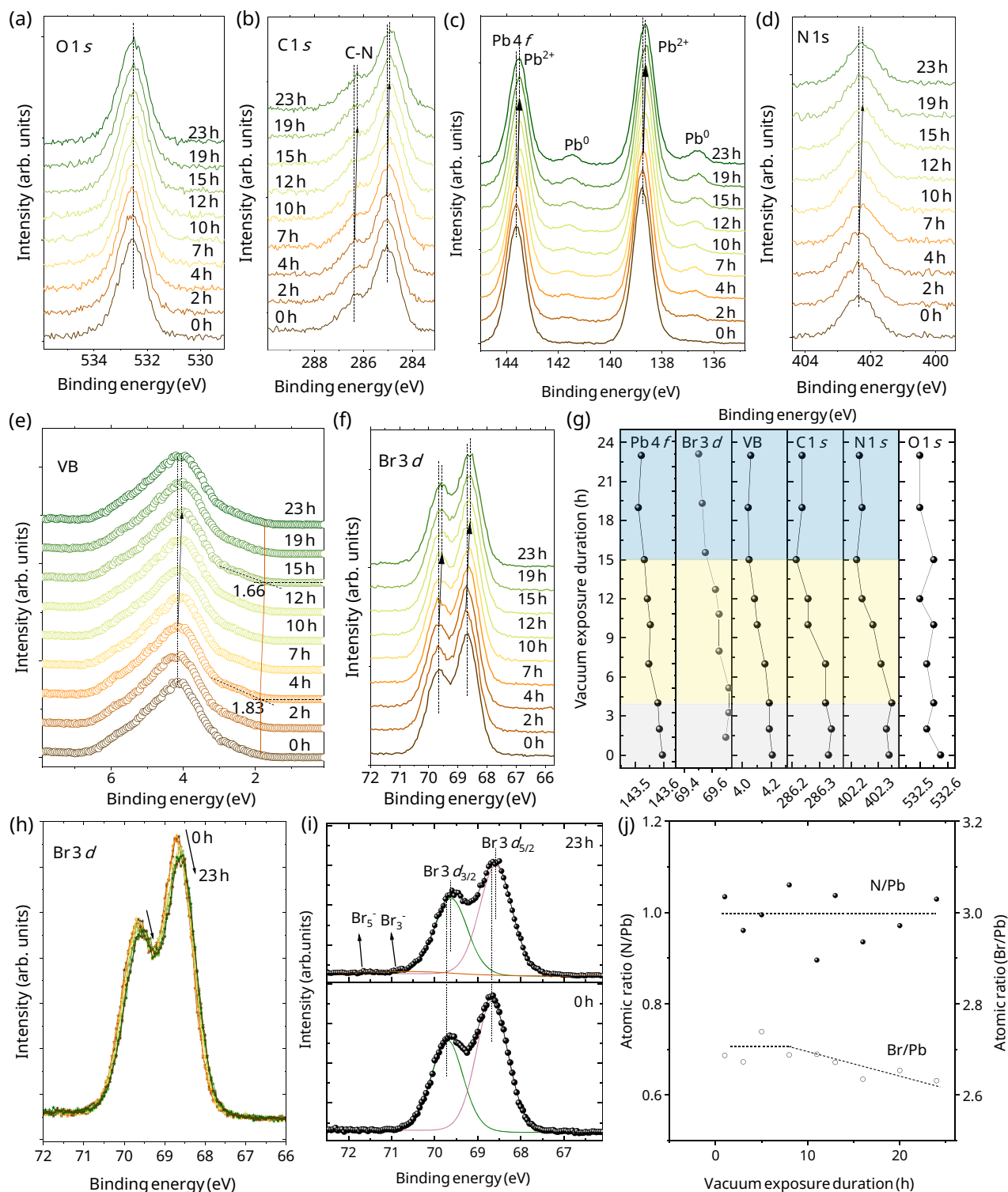


Fig. 3 | Evolution of MAPbBr₃ chemical state with vacuum exposure. a–f In-situ O 1s, C 1s, Pb 4f, N 1s, VB and Br 3d XPS spectra measured on MAPbBr₃ surface after different vacuum exposure duration under pressure of 1.6×10^{-9} mbar; **g** Peak position versus vacuum exposure duration for various elements, in which frames with different peak shift trends are shown in grey, yellow and blue backgrounds, respectively; **h** Variation of Br 3d intensity with vacuum exposure, indicating the vacuum-induced formation and desorption of intermediate molecular bromine species in ultrahigh vacuum condition. **i** Comparison of Br 3d XPS spectra before

and after 23 h vacuum exposure, which highlights the appearance of oxidized peaks after vacuum exposure as labelled by black arrows, suggesting the vacuum-induced formation of oxidized bromine species such as trimer (Br₃⁻). **j** Variation of N/Pb and Br/Pb atomic ratio with vacuum exposure, where the constant value of former excludes the decomposition of perovskites, while the latter confirms the ultrahigh-vacuum-induced desorption of molecular bromine species. Source data are provided as a Source Data file.

opposite process of water adsorption has occurred during vacuum exposure and causes all the peaks except O 1s to shift towards lower BE. (2) In Pb 4f XPS spectra, small Pb⁰ shoulders start emerging near Pb²⁺ peaks from 2 h and increase continuously^{26,32}. The Pb⁰ formation is caused by X-ray-induced reduction of Pb²⁺ during the continuous measurements conducted on the same spot, which has been studied by us and others previously^{32–34}. This non-synchronous evolution between Pb⁰ formation and peak shift observed in Fig. 3g suggests that a different mechanism is causing the peak shift rather than X-ray irradiation itself. (3) No obvious change is observed in the intensity of Pb 4f, N 1s as well as C–N peak of C 1s spectra as shown by the non-stacked line graph of XPS spectra in Supplementary Fig. 16. This result indicates that no obvious decomposition of perovskite occurs under vacuum exposure^{35,36}. Meanwhile, a drop of C–C peak is observed due to the vacuum-induced removal of organic contaminations. (4) Small features appear at higher BE in the Br 3d XPS spectra at a later stage of vacuum exposure, which correspond to oxidized species of bromine such as Br₃[–] (Fig. 3i). The intensity of the oxidized bromine peak is much more pronounced if the long vacuum exposure is performed under a moderate vacuum with pressure of 1×10^{-6} mbar after which the sample is loaded to XPS chamber for measurement. (Supplementary Fig. 17) This contrast in oxidized bromine peak intensity arises probably because some intermediate molecular bromine species (i.e., Br₂) desorb in the ultrahigh vacuum of the XPS chamber (pressure = 1.6×10^{-9} mbar). The removal of intermediate bromine species can be evidenced by the decrease in Br peak intensity after long vacuum exposure. (Fig. 3h, j) These observations suggest that trimer (Br₃[–]) is generated by the binding of intermediate bromine species (i.e., Br₂) with a third Br[–] ion.

The trimer structure of bromine (Br₃[–]) in the perovskite lattice is equivalent to a Br⁺ interstitial bound by two Br[–] on both sides, like the I₃[–] in iodide perovskite^{38–40}. This Br⁺ interstitial can cause slight lattice expansion to MAPbBr₃⁴¹, and also create a deep electron trap in the bandgap of MAPbBr₃ as reported by the theoretical calculations in an earlier work⁴². Therefore, one can speculate that the re-occurrence of lattice expansion observed in XRD and the emergence of emission at 2.05 eV observed in PL after long vacuum exposure are due to the formation of bromine trimers. This defect assignment is also consolidated by the ultralong vacuum-induced emission decay in Fig. 1h, since the carrier de-trapping from positively charged halogen interstitial is known to involve a significant geometrical relaxation and the corresponding emission has a long decay lifetime³⁸.

Generalization of structural evolution trend under vacuum exposure

Further, we investigate the structural evolution of perovskites with different cation and halogen (i.e., FAPbI₃, CsPbBr₃, Cs_{0.05}MA_{0.1}FA_{0.85}PbI₃) with pre-adsorbed water under vacuum exposure, to verify whether the same vacuum-induced structural evolution trend applies. The experimental details for the synthesis of FAPbI₃ and Cs_{0.05}MA_{0.1}FA_{0.85}PbI₃ polycrystalline film, and CsPbBr₃ powder are included in the Methods section.

XRD analysis in Supplementary Fig. 18 reveals that the FAPbI₃ film undergoes structural evolution similar to that of MAPbBr₃ during vacuum exposure. The diffraction peaks first exhibit a shift towards higher angle after short exposure (1×10^{-6} mbar, 5–10 h), and reverse backwards after longer exposure (24 h). This result indicates that vacuum exposure induces water desorption and lattice shrinkage, and further leads to the formation of iodine trimers and slight lattice expansion in the FA-based perovskite as well.

Similarly, the Cs_{0.05}MA_{0.1}FA_{0.85}PbI₃ film with pre-adsorbed water exhibits similar evolution trend during vacuum exposure. The as-prepared film shows a polycrystalline nature with crystal structure matching the cubic phase at room temperature (Fig. 4a, b). Figure 4c–e & Supplementary Fig. 19 present the PL contours of

Cs_{0.05}MA_{0.1}FA_{0.85}PbI₃ film after different vacuum exposure duration (1×10^{-6} mbar, 0 h, 5 h, 16 h, 24 h). Figure 4f plots its emission peak position in the temperature range of 20–200 K after different vacuum exposure duration. Initially at 0 h, the emission peak position displays two sudden changes at -170 K and -60 K. The sudden PL peak position change with temperature corresponds to the phase transition of perovskite. We perform varying-temperature XRD on Cs_{0.05}MA_{0.1}FA_{0.85}PbI₃ film to study its phase transition, for which the results are reported in Supplementary Fig. 20. The cubic α phase observed at high temperature (i.e., 300 K) first transits towards the tetragonal phase when the temperature decreases below -260 K⁴³. With the temperature decreasing further, a second structural change occurs at -170 K, which can be ascribed to the transition between two tetragonal phases, referring to the earlier works on FA_xMA_{1-x}PbI₃^{44,45}. Due to the limited temperature accessible in our XRD instrument, the further structural evolution below 100 K is not investigated. We speculate that the abrupt PL peak position change at -60 K corresponds to the transition between the tetragonal and orthorhombic phases.

After 16 h vacuum exposure, the sudden change at -170 K observed in the PL contour remains, while the one at -60 K is modified. This result suggests that the phase transition toward orthorhombic phase is modified due to vacuum exposure-induced water desorption and elimination of lattice expansion. With even longer vacuum exposure (i.e., 24 h), the phase transition again recovers, which is similar to the case of MAPbBr₃.

Besides, the low temperature PL analysis of Cs_{0.05}MA_{0.1}FA_{0.85}PbI₃ film shows that its emission peak first redshifts with vacuum exposure, which reverses during further vacuum exposure. (Fig. 4g) The reversal of PL peak position suggests re-occurrence of lattice expansion which probably originates from the formation of iodine trimer (I₃[–]). The presence of iodine trimers is verified by the XPS measurement of a film after 24 h of vacuum exposure. Specifically, a peak corresponding to oxidized iodine species is observed at the higher BE side of I[–] in the I 3d XPS spectra. (Fig. 4h) Supplementary Fig. 21 also presents the C 1s, N 1s and Pb 4f XPS spectra of the film. Further, the presence of iodine trimers and the intermediate iodine dimers is also verified by the UV-vis absorption spectra shown in Fig. 4i, which is obtained by dipping a heavily-exposed sample inside toluene for several hours and measuring absorption of the solvent. The above results of luminescence and compositional evolution imply that Cs_{0.05}MA_{0.1}FA_{0.85}PbI₃ exhibits a structural evolution trend similar to MAPbBr₃ during the 24 h vacuum exposure. A further extension of the vacuum exposure duration up to 40 h induces the generation of FAPbI₃ as shown in Supplementary Fig. 22. Due to the smaller bandgap of FAPbI₃, the low temperature PL shows a small redshift in the peak position from 24 h to 40 h as seen in Fig. 4g.

In addition, we investigate the structural and luminescence evolution of fully inorganic perovskite (CsPbBr₃) with pre-adsorbed water during vacuum exposure (1×10^{-6} mbar). Supplementary Fig. 23 reports the XRD evolution of CsPbBr₃, in which the diffraction peak also shifts toward a higher angle upon vacuum exposure, indicating the recovery of lattice expansion. However, no re-occurrence of lattice expansion is observed in the later stage of vacuum exposure, indicating no formation of bromine trimer. This is evidenced further by the PL and XPS results in Supplementary Figs. 24 and 25. Specifically, the low-temperature PL spectra of CsPbBr₃ always show a two-peak feature, and no extra sub-bandgap emission appears after long vacuum exposure. Besides, the Br 3d XPS spectra of CsPbBr₃ after long vacuum exposure do not show oxidized species of bromine. Note that the feature observed at a slightly higher BE of Br 3d XPS spectra corresponds to the residue of PbBr₂ precursors.

Atomistic model, DFT calculations and Raman spectroscopy

Our experiments suggest that lattice expansion, through water adsorption at the surface lattice, inhibits the formation of halogen

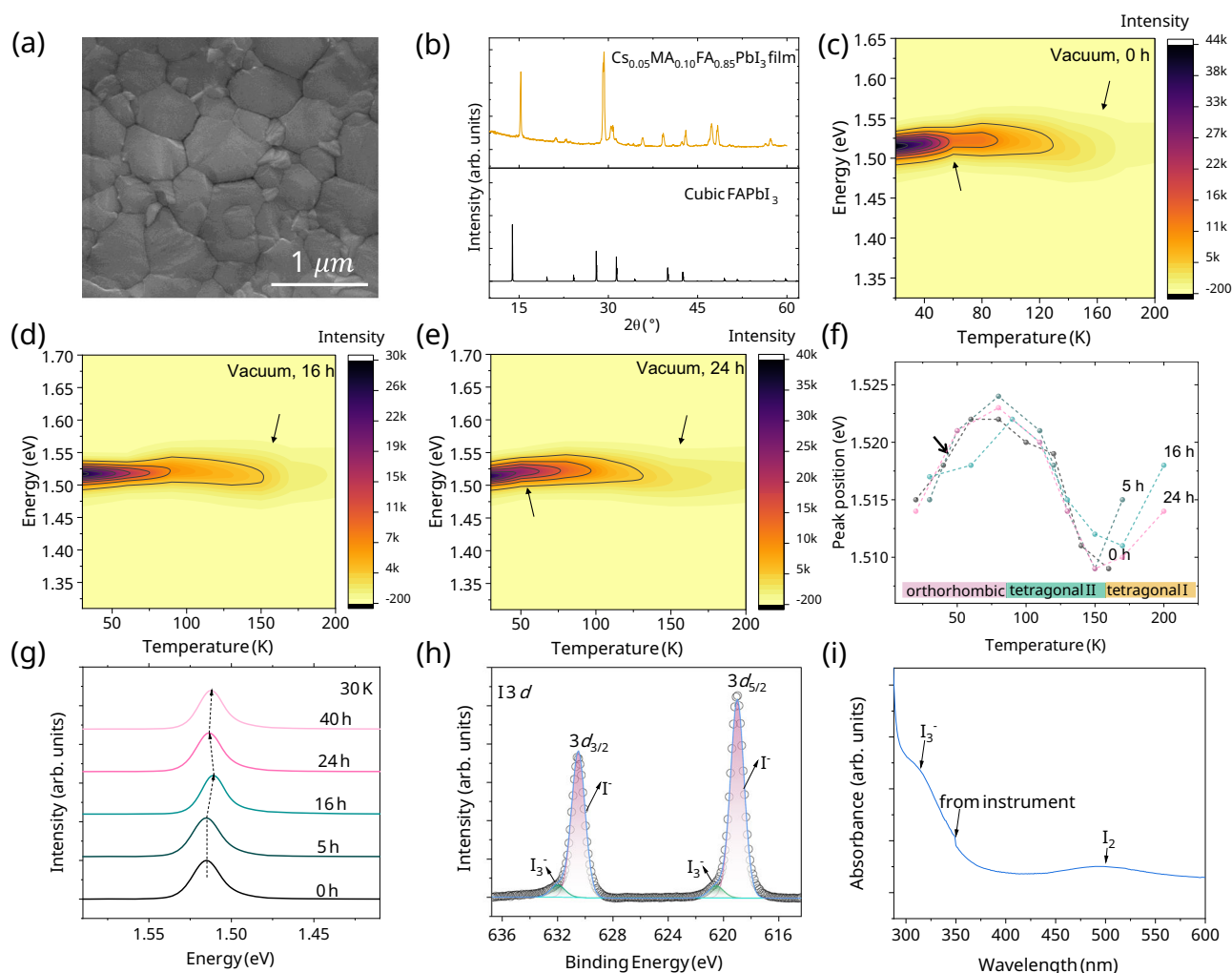


Fig. 4 | Characterization of as-synthesized $\text{Cs}_{0.05}\text{MA}_{0.1}\text{FA}_{0.85}\text{PbI}_3$ film and vacuum-induced structural evolution. **a** SEM image and **b** XRD of $\text{Cs}_{0.05}\text{MA}_{0.1}\text{FA}_{0.85}\text{PbI}_3$ film, showing a larger lattice constant in comparison with the reference pattern of cubic FAPbI_3 ; PL contours of $\text{Cs}_{0.05}\text{MA}_{0.1}\text{FA}_{0.85}\text{PbI}_3$ film after **c** 0 h, **d** 16 h, and **e** 24 h vacuum exposure under pressure of 1×10^{-6} mbar, in which two sudden changes of peak position are labelled at -170 K and -60 K by the black arrows. **f** PL peak position versus temperature after different vacuum exposure duration, the inset shows the phase transition at the corresponding temperature.

g Normalized 30 K PL spectra after different vacuum exposure durations. **h** $13d$ XPS spectra after vacuum exposure for 24 h, in which the Γ^- peak and oxidized peak are shown in purple and green, respectively; **i** UV-vis absorption spectra of the solvent with dissolved I_2 and I_3^- from the 24 h vacuum-exposed film, showing absorption features at 500 nm and 320 nm, respectively, as labelled by the black arrows. The other absorption feature at 350 nm is coming from the instrument. Source data are provided as a Source Data file.

trimers (X_3^-) in the hybrid perovskite material. On the other hand, formation of halogen trimers in perovskite crystals has been thoroughly reported in the literature. The formation of I_2 and/or Br_2 in perovskites is typically regarded as caused by factors such as light irradiation^{46–48}, but due to our experimental conditions, we can regard the effect of light or other extrinsic factors as irrelevant. We instead hypothesize that halogen trimers (i.e., Br_3^-) in our system are formed spontaneously due to thermal effects related to the removal of lattice expansion. In particular, the appearance of a metastable phase (“60° phase”), close in energy to the ground state (“0° phase”), characterized by a MA^+ molecule rotation of 60° with respect to the ground state configuration, plays a critical role. This 60° phase presents a shorter Br-Br distance, which may facilitate the formation of Br_2 molecules at the first stage, and then of trimers, by bonding with a third Br⁻ ion. The potential role of this 60° phase as a precursor for Br_2 formation was outlined for a model material of MAPbI_3 in ref. 49, but it is not clear what factors might stabilize the 60° phase, whether this phase can be detected experimentally, and if a refined understanding of the

strengths of the bonds involved in the process can be achieved. In the following, we present a model to try to answer these questions in combination with our experimental results.

Knowing that water adsorption at the surface expands the lattice, we explore the relative stability of the 60° phase against the 0° phase by means of DFT calculations. As a starting point, we select the structure in refs. 50,51, and relax it, obtaining the ground state of the system. At room temperature, the temperature fluctuations cause a semi-random arrangement of the MA^+ molecule, and the structure adopts a cubic phase with $Pm\bar{3}m$ space group (No. 221). We first verify the stability of the 60° phase by fully relaxing the structure after setting the angle, and calculating Γ point phonons. All phonon frequencies at Γ are real, and the structure is very close in energy to the relaxed ground state (3 meV). We thus treat it as a metastable minimum. We choose a rather strict force threshold (see Methods section). We also note that, when starting the relaxation with MA^+ rotation angles close to 0° or 60°, the structure tends to relax towards the closest one of those two. We subsequently explore the stability of the

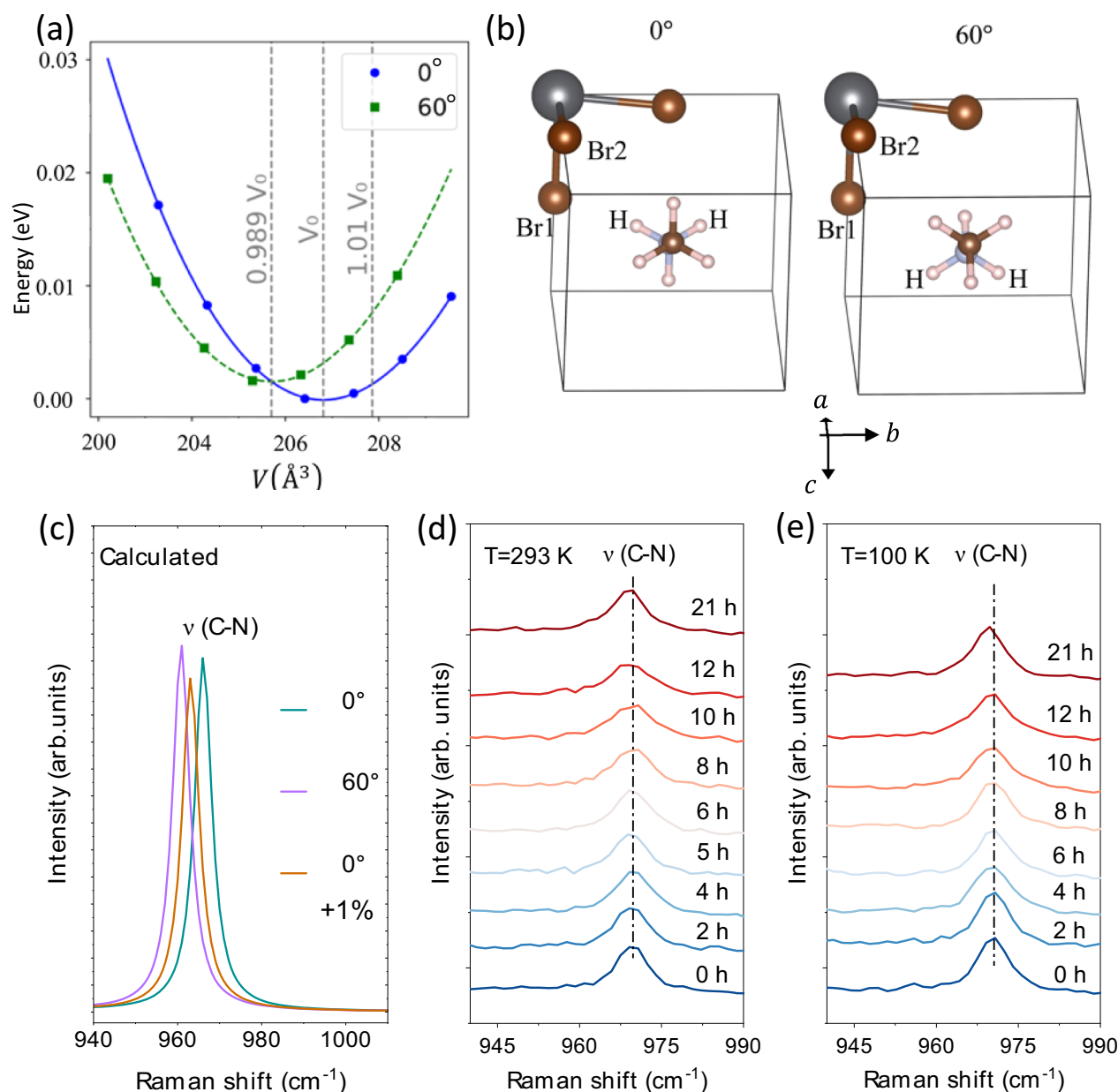


Fig. 5 | The recognition of a metastable phase (“60° phase”) in MAPbBr₃ lattice lacking expansion. **a** Fitted energy vs volume curves of the 0° and 60° phases. Blue (green) circles (squares) correspond to the 0° (60°) phase. **b** The unit cell of the relaxed 0° (left) and 60° (right) structures. The most relevant Br and H atoms for bonding analysis are highlighted with letters. **c** Calculated Raman spectra of C–N

stretching mode for MAPbBr₃ at ~966 cm⁻¹ with the 0° phase, 60° phase and 0° (+1%) phase overlaid. The in-situ experimental Raman spectra in wavenumber range of 940–990 cm⁻¹ after vacuum exposure under pressure of 1×10^{-4} mbar for different duration showing the peak of C–N stretching mode at **d** 293 K and **e** 100 K. Source data are provided as a Source Data file.

two phases as volume is changed above and below the ground state volume, by performing fixed volume, variable cell structural relaxations. After fitting the energy vs volume curves of both phases to a third-order Birch–Murnaghan model^{52,53}, the results shown in Fig. 5a are obtained: as expected, the 0° ground state is significantly more stable at a volume increment as modest as 1%, where the 60° phase is over 6 meV higher in energy. This strongly supports the suppression of the 60° phase when water is adsorbed. On the other hand, a volume reduction of 1.01% is enough to render the 60° phase more stable. Thus, slight changes in strain or temperature fluctuations are expected to allow the presence of the 60° phase in some regions once the water is desorbed or where it was never present. In Supplementary Table 4 of the SI, we provide the fitted parameters of the curves. The result suggests that certain regions across the material under strain gradients and/or temperature fluctuations may experience a high probability of

presenting the 60° arrangement when the lattice has no water incorporated. In contrast, when water is present, the concomitant lattice expansion significantly decreases the probability of the 60° phase appearing in the crystal, favoring a higher stability of the material.

We next try to find a fingerprint of the 60° structure in our vacuum experiments through Raman spectroscopy, as we will detail below. These measurements are combined with DFT calculations of the Raman spectra of three selected phases: 0° phase (ground state), 60° phase, and 0° structure relaxed at fixed volume, where the volume is 1% larger than the ground state volume. This latter structure is used to simulate the system with water adsorbed. The full calculated spectra are shown in the SI, Supplementary Fig. 30. Figure 5c shows the Raman spectra of MAPbBr₃ calculated by DFT for the three phases, overlaid and zoomed into a frequency range that captures the characteristic C–N stretching mode at ~966 cm⁻¹, consistent with the reported results

in ref. 54. We see that the 0° phase peak corresponds to the blue shifts with respect to the $+1\%$ volume phase, while the 60° one redshifts. We perform an in-situ Raman measurement in vacuum with a pressure of 1×10^{-4} mbar. The water desorption can occur under this vacuum condition. Supplementary Fig. 26 shows the experimental Raman spectra in the range of $850 - 1750 \text{ cm}^{-1}$ captured after different vacuum exposure duration at 293 K and 100 K. As the magnified spectra show in Fig. 5d, e, the C–N stretching mode of the MA^+ molecule ($\nu(\text{C}-\text{N})$) shows a visible redshift in peak position with vacuum exposure. We observe that the redshift expected by the calculations is present in the in-situ Raman experiment, which strengthens the case in favor of the increase in the presence of the 60° phase in the crystal as the vacuum exposure time increases. C–N stretching peak position also exhibits a sudden shift in the tetragonal-orthorhombic phase transition as demonstrated in a previous study⁵⁴. This suggests that the metastable phase formed due to lack of lattice expansion probably results in the change of the degrees of freedom and modifies the phase transition as observed in our PL study.

The event of Br_2 formation in the lattice is presumably preceded by a series of thermally induced bond-breaking events, so the two Br atoms are released and then attached to each other. We have computed the strengths of the bonds that the two closest Br atoms in the lattice (Br1 and Br2) form with their neighboring atoms within a maximum distance of 3.3 Å, in the three structures listed in the previous paragraph. The atoms Br1 and Br2 are shown in Fig. 5b, along with the Pb atom in the unit cell and the 2 strongest bonding H atoms. In the SI Supplementary Table 3 we show all the calculated ICOHP parameters^{55–57}. The cumulative bond strength on atoms Br1 and Br2 is shown in Table 1. The decrease in cumulative bond strength on the Br1 atom in the 60° phase, coupled to the decrease of the Br–Br distance, is interpreted as an increased ability of the Br atoms in the 60° phase to detach from their position due to thermal fluctuations, in comparison to the other two phases. Meanwhile, the change from the expanded volume phase to the 0° phase is comparatively modest, or even detrimental. The cumulative bond strength on the atom Br2 in the 60° phase is smaller than that of the Br1 atom in the other two phases, indicating Br2 is the bottleneck for detachment of both atoms to form a Br_2 dimer. This reinforces the view of the 60° phase-mediated halogen dimer formation, as described schematically in Supplementary Fig. 27. In Supplementary Fig. 31 we show in detail the bonds we have used for our calculations.

Based on the above-proposed mechanism, the interior perovskite lattice should be intrinsically characterized with trimers due to the lack of lattice expansion. We perform secondary ion mass spectroscopy (SIMS) measurements on a MAPbBr_3 crystal after 40 h vacuum exposure (1×10^{-6} mbar). Ar Cluster ion gun with an energy of 10 keV is used for the SIMS measurements to minimize the ion-beam-induced damage. Supplementary Fig. 28 reports the SIMS results in which the secondary ions are collected in a negative ion mode in a sputtering time of 2200 s, which confirms the presence of Br_2 and Br_3^- species along the full depth of measurement. According to the profile measurement of the sputtered surface, the full sputtered depth in the sputtering duration of 2200 s is about $\sim 4.41 \mu\text{m}$. Besides, the

depth distribution of Br_3^- species shows that a slightly higher density is detected at surface, which decreases with depth and becomes stable at $\sim 1.10 \mu\text{m}$. The slightly higher trimer density at surface is likely due to the presence of surface defects such as bromine vacancies. We have also calculated the adsorption energy of Br_2 on MA-Br -terminated MAPbBr_3 slabs, getting an energy of -0.99 eV that is favorable for adsorption. See the calculation details in Supplementary Fig. 32 in the SI, in which we display the relaxed configuration of the slab with the molecule. Thus, in our study, a vacuum with a pressure of 1×10^{-6} mbar does not cause significant desorption of Br_2 species, while an ultrahigh vacuum of 1.6×10^{-9} mbar enables its desorption to occur as shown in XPS analysis.

On the other hand, SIMS results show that the Br_3^- species are distributed uniformly in the lateral dimension as indicated by the 2D and 3D maps in Supplementary Fig. 28. We also detect abundant PbBr_3 species along the depth, indicating that the trimer possesses the same structure as reported in the previous study³⁸, where a bromine interstitial is bound by two bromine atoms of the lattice sites. The uniformity of Br_3^- in the interior lattice is also evidenced by low-temperature confocal PL map. Supplementary Fig. 29 presents the confocal PL map of MAPbBr_3 in the emission range of $2.0 - 2.1 \text{ eV}$ with a spatial resolution of 200 nm, which is acquired on the inner surface of an $\sim 20 \mu\text{m}$ thick layer cracked from a single crystal.

Implications for the photovoltaic performance

It is important to investigate the impact of water adsorption-desorption-induced lattice expansion-recovery on solar cell device performances. To probe the effect of lattice expansion, we synthesize three $\text{Cs}_{0.05}\text{MA}_{0.1}\text{FA}_{0.85}\text{PbI}_3$ films in nitrogen and subsequently expose the films to air (RH = 70%) for 5, 15, and 35 min, which are named as sample A, B, C, respectively. The three films are used to assemble solar cells with a structure shown in the inset of Fig. 6a, for which the synthesis detail is described in the Methods section. The recorded J–V curves of three solar cells are reported in Fig. 6a. The boxplot of power conversion efficiency (PCE) is presented in Fig. 6b. The results demonstrate that the incorporation of water molecule in the perovskite film can improve the PCE of the solar cell. The boosted PCE can be attributed to water-induced lattice expansion and consequently inhibited non-radiative recombination as revealed in earlier works^{10,12}.

Further, we prepare four perovskite films, which are all exposed to air for 35 min before further vacuum exposure for different durations (1×10^{-6} mbar, 1 h, 8 h, 16 h, 24 h). The minimum vacuum exposure time is 1 h, which is mandatory for depositing the subsequent electron transport layer. The J–V curves are measured and compared for four solar cells assembled using the four films. As shown in Fig. 6b, c, compared to the solar cell made with 1 h exposure (PCE = 21.33%), the efficiency decreases slightly for the one with 8 h vacuum exposure (PCE = 20.85%). With further vacuum exposure, the efficiency decreases further (PCE = 19.17% for 16 h, and 17.61% for 24 h). The drop of PCE indicates that as the vacuum gradually eliminates the pre-adsorbed water, the lattice lacks expansion and iodide trimer-related defects are generated, which cause the reduction of PCE, as expected.

The stability of the four solar cells is explored by measuring the J–V curves after 1 month of storage in nitrogen. The boxplot in Fig. 6d shows that the PCE of solar cells with 1 h and 8 h vacuum exposure remains to have 98.7% and 95.6% of the initial value. By contrast, the PCE of solar cells with 16 h vacuum exposure is only 88.6% of the initial value. This decline verifies that the vacuum-induced removal of water and elimination of lattice expansion leads to further generation process of trimers and deterioration of the device performance during storage. For the solar cell with 24 h vacuum exposure, the PCE remains to be 95.7% of the initial value. This indicates that the iodine trimers have already been formed during the prolonged vacuum exposure, and no obvious structural degradation occurs during the storage in a nitrogen atmosphere.

Table 1 | In the first two rows, cumulative ICOHP values of bonds involving atoms Br1 and Br2 in the three considered structures

	$0^\circ \text{ V} + 1\%$	0°	60°
ICOHP-Br1 (eV)	−4.19	−4.37	−3.94
ICOHP-Br2 (eV)	−3.54	−3.69	−4.17
<i>d</i> (Br1-Br2) (Å)	3.86	3.80	3.72

In the third row, the distance between atoms Br1 and Br2.

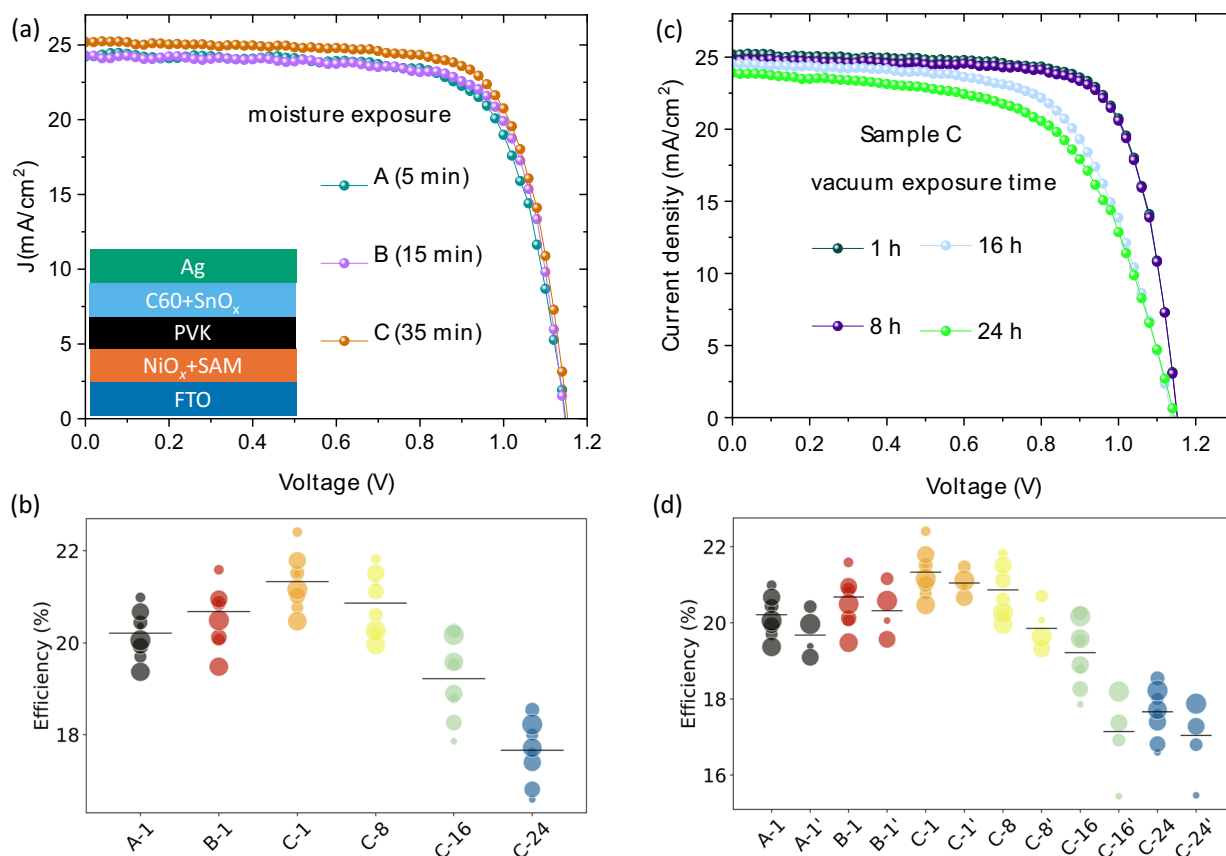


Fig. 6 | Effect of moisture exposure and subsequent vacuum exposure on $\text{Cs}_{0.05}\text{MA}_{0.1}\text{FA}_{0.85}\text{PbI}_3$ film-based solar cell. **a** J–V curves of three solar cells assembled with three films after exposure to air (RH = 70%) for 5 mins (sample A), 15 mins (sample B), 35 mins (sample C), respectively, inset shows the structure of solar cell. **b** Boxplot showing the PCE of solar cells presented in this figure. **c** J–V

curves of four solar cells assembled with film C after exposure to vacuum (pressure = 1×10^{-6} mbar) for 1 h, 8 h, 16 h, and 24 h, respectively. **d** Boxplot showing the variation of PCE for solar cells in (a, c) before and after 1 month storage in glove box, in which the cells after 1 month storage are labelled by adding a prime ('). Source data are provided as a Source Data file.

In summary, we uncover the necessity of a slight lattice expansion for the stability of hybrid perovskite. The slight lattice expansion induced by water adsorption can avoid the formation of a considerable number of bad defects, such as halogen trimers in perovskite, and increase the efficiency of solar cell devices. On the contrary, upon the desorption of water through external factors such as vacuum exposure, the lattice expansion recovers, which promotes the formation of a halogen dimer molecule. This structural change paves the way for the further formation of halogen trimer in the perovskite lattice, which is equivalent to a positively charged halogen interstitial, detrimental to solar cell devices. This work suggests that a mild lattice expansion of perovskite can increase the stability of perovskite and relevant devices. In the assembly process of perovskite devices, the water molecules adsorbed at perovskite surface can be removed by the vacuum exposure during subsequent processes such as vacuum exposure. The defect generation under vacuum exposure constitutes one of the carrier loss channels in the devices. This work therefore lends inspiring insights to the optimization of perovskite-based devices for space applications. Besides, our study also provides useful guidance for accurate characterization of perovskites using vacuum-based techniques, which is crucial for understanding the basic properties of this class of material and further boosting the development of perovskite-based optoelectronics.

Methods

MAPbBr₃ single crystal preparation

MAPbBr₃ single crystal is synthesized based on the inverse temperature crystallization method. 111.9 mg MABr and 367 mg PbBr₂

are first dissolved in a mixed solvent of 1 mL DMF and stirred for 7 h. The precursor solution is subsequently transferred to a glass vial and heated in an oil bath. The temperature is increased from 60 °C to 80 °C at a rate of 3 °C/h for crystal nucleation. After the seed appears, it is kept at 80 °C for 3 h for crystal growth. The grown crystals are finally taken out from the vial and dried, which are exposed to ambient conditions (room temperature, 1 atm, RH = 70%) for 3 h before being transferred to the vacuum chamber for further investigation.

MAPbBr₃ film preparation

ITO glass is cleaned thoroughly and treated with ozone for 10 min. 1 mmol PbBr₂ and MABr are stirred at room temperature and dissolved in the mixed solvent of DMF and DMSO in a volume ratio of 1:1. 50 μL precursor is filtered and spin coated on the substrate, which is annealed at 110 °C for 15 min. A thermal flow is applied during the spin-coating process.

$\text{Cs}_{0.05}\text{MA}_{0.1}\text{FA}_{0.85}\text{PbI}_3$ film preparation

19.5 mg CsI, 23.9 mg MAI, 219.3 mg FAI and 746.8 mg PbI₂ are weighed and placed in 1 mL mixed solvent of DMF and DMSO with a volume ratio of 4:1 in a glove box. 15.2 mg MAI is also added to improve the crystallinity of the synthesized film. The solution is then stirred for 6 h to obtain a fully dissolved precursor. In parallel, a glass slide is cleaned thoroughly, treated with ozone and transferred to the glove box. The precursor solution is dropped onto the glass slide, which is first spun at 1000 rpm with an acceleration speed of 500 rpm/s for 10 s and then spun at 5000 rpm for 30 s. At the last eight seconds of spin-coating,

120 μL anti-solvent (anisole) is added onto the glass. The film is finally obtained after an annealing at 100 °C for 20 min.

CsPbBr₃ powder synthesis

2.13 g CsBr and 4.04 g PbBr₂ are weighed and dissolved in 10 mL DMF by stirring; 400 μL precursor solution is taken out, and 1 mL toluene is subsequently added to the precursor solution. CsPbBr₃ microcrystals are obtained after reaction and centrifugation.

Solar cell fabrication and characterization

FTO glass with $1.5 \times 1.5 \text{ cm}^2$ is cleaned thoroughly, dried and treated by ozone. A NiO_x solution is spin-coated onto the FTO glass, which is then heated at 150 °C for 10 min. Further, 0.5 mg/mL Me-4PACz solution is spin-coated and annealed for 10 mins at 100 °C. Subsequently, the Cs_{0.05}MA_{0.1}FA_{0.85}PbI₃ film is prepared using the same precursor as the above film synthesis, but with a different crystallization method: after the precursor is spin-coated onto the substrate, the solvent is removed by a short treatment (~20 s) under low vacuum (1×10^{-3} mbar) instead of anti-solvent method. The obtained perovskite film is passivated first by 1 mg/mL m-f-PEAI and annealed at 100 °C for 5 mins and later by 2 mg/mL PI annealed at 100 °C for 1 min. The perovskite film is subjected to different vacuum treatment time, and afterwards, 20 nm C60 is deposited by thermal evaporation. Following this process, 20 nm SnO₂ is deposited by atomic layer deposition. Finally, a 100 nm thick Ag is coated by thermal evaporation, giving a solar cell with p-i-n structure of active area of 0.07 cm². The photovoltaic performance is measured with a source meter (Keithley 2400) with 100 mA/cm² illumination (AM 1.5 G) and a calibrated silicon cell as reference (Enlitech, KG2).

Structure, composition, and morphology characterization

XRD of the samples is measured using a Bruker D8 ADVANCE diffractometer. The samples are placed in a holder, and the sample surface is kept at the same height as the sample holder, which is located at the focusing point of the X-ray. XPS is conducted on ESCALAB Xi+. The pressure of the XPS chamber is 1.6×10^{-9} mbar, and the in-situ XPS measurement is carried out on the same crystal after every two hours. The crystal is wrapped in an In foil to avoid charging during XPS measurements. Raman spectra are captured using Renishaw, inVia Qontor under excitation light of 785 nm. The pressure of the Raman chamber is 1×10^{-4} mbar, and the in situ Raman measurement is conducted every few hours continuously. FTIR spectra are acquired on air-exposed powder obtained from the grinding of a single crystal using BRUKER INVENIOR. SIMS measurement is performed on 40 h vacuum-exposed crystal using TOF-5 (GmbH) with Ar-Cluster gun (CGIB) of 10 keV in negative ion mode.

Optical characterization

Temperature-dependent PL measurements are conducted using a thermostat (CS202AE-DMX-1AL, ARS, USA) with helium cooling, which is assembled with 385 nm LED for excitation and a PL spectrometer (QYAS, BiaoQi optoelectronics, China). The pressure of the cryostat chamber is 1×10^{-6} mbar, and the in-situ PL measurement is carried out on the same crystal after different durations. The incident light is switched on only during the PL capturing to avoid the light-induced structural change. An optical spectrometer (Edinburgh Instruments FLS1000) is used for conducting the TRPL measurements. The absorption spectra are measured in transmission mode using a UV-Vis spectrophotometer (UV1920-100, Prism, China). For confocal PL measurement, a piece of the surface layer is exfoliated from the surface of MAPbBr₃ single crystal and loaded onto the copper grid. The grid is cooled to 77 K, and the confocal PL map is acquired on the inner surface of the layer under 405 nm excitation.

DFT calculations

DFT calculations are performed using VASP code^{58,59}, version 5.4.4. Interactions of electrons with ion cores are represented using projected atomic wave formalism⁶⁰, and the exchange-correlation functional is parameterized with the nonlocal vdW exchange-correlation functional optB86b-vdW⁶¹ VDW. For full unit cell relaxations of the 0° and 60° systems, as well as for their fixed volume relaxation at different volumes, we use a $6 \times 6 \times 6$ monkhurst-pack grid and a force threshold of 10^{-4} eV/Å. ASE⁶² is used for structure manipulation; VESTA⁶³ for structure visualization; Phonopy⁶⁴ package, combined with tools available online^{65,66} for phonon and Raman calculations; and LOBSTER^{55–57} for bond strength calculation. See additional details in the SI.

Data availability

The data that support the findings of this study are available in the Supplementary Information and Source Data file. Source data are provided with this paper. The source data generated in this study have been deposited in the Figshare database⁶⁷ with DOI of <https://doi.org/10.6084/m9.figshare.29618630>.

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Author contributions

T.L. performed the experimental studies including the crystal synthesis and vacuum-induced structural and optical evolution. Y.Li, X.R., Z.Z., Z.C., Z.L., and Y.H. assisted the experimental studies. Y. Luo carried out the solar cell fabrication and analysis. U.P. performed the computational studies. J.X. and J.Y. participated in the discussion and gave valuable suggestions. Z.D. supervised the work and wrote the manuscript. P.G. provided financial support and gave valuable suggestions.

Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to Zhiya Dang, Urko Petralanda or Pingqi Gao.

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